

Multiple Choice

1. **Answer:** C

Options: A) $a, b \in H \Rightarrow a * b \in H$; B) $a \in H \Rightarrow a^{-1} \in H$; C) $a, b \in H \Rightarrow a * b^{-1} \in H$; D) H contains the identity element

Note: Correct option : C - $a, b \in H \Rightarrow a * b^{-1} \in H$

1. **Answer:** B

Options: A) 3; B) π ; C) 3π ; D) $3/2$

Note: Correct option: B - π

2. **Answer:** C

Options: A) 27; B) 28; C) 35; D) 37

Note: Correct option: C - 35

3. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

4. **Answer:** A

Options: A) $2x^2 + 5$; B) $6x^2 + 4x + 6$; C) 0; D) $x^2 + 1$

Note: Correct option: A - $2x^2 + 5$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(1, 6)'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-1'.

Long Form Response

11. **Answer:** II only. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.